SAFETY DATA SHEET

ECOLAB

ECO-CLEAN ELITE 500 HD DEGREASER

1. Chemical product and company identification

Chinese name : 除油劑

Product name : ECO-CLEAN ELITE 500 HD DEGREASER

Other names : None.

Product code : 903372

Recommended use and restrictions : Degreaser

Use only for the purpose on the product label.

Manufacturer/Supplier Address/Telephone : Ecolab Limited
6F, No. 13, Sec. 2, Peitou Rd., Peitou Dist.,
Taipei, 112, Taiwan, R.O.C.
(02) 2898-4800

Emergency telephone number : 1-651-222-5352 (in USA)

2. Hazards identification

GHS Classification : CORROSIVE TO METALS - Category 1
SKIN CORROSION/IRRITATION - Category 1A
SERIOUS EYE DAMAGE/ EYE IRRITATION - Category 1

GHS label elements

Symbol :

Signal word : Danger

Hazard statements : May be corrosive to metals.
Causes severe skin burns and eye damage.

Precautionary statements

Prevention : Wear protective gloves. Wear eye or face protection. Wear protective clothing. Keep only in original container. Wash hands thoroughly after handling.

Response : IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or physician. IF SWALLOWED: Immediately call a POISON CENTER or physician. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. Immediately call a POISON CENTER or physician. IF IN EYES: Immediately call a POISON CENTER or physician.

Other hazards : None known.

3. Composition/information on ingredients

Nature of material : Mixture

Chemical properties : Degreaser

Hazardous ingredients	Concentration Range (%)	CAS number
sodium hydroxide	5 - 20	1310-73-2
triethanolamine	1 - 5	102-71-6
2,2'-iminodiethanol	< 1.0	111-42-2

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

4. First aid measures

First aid measures

Eye contact : In case of contact, immediately flush eyes with cool running water. Remove contact lenses and continue flushing with plenty of water for at least 15 minutes. Get medical attention immediately.

Skin contact : In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Get medical attention immediately. Wash clothing before reuse. Clean shoes thoroughly before reuse.

Inhalation : If inhaled, remove to fresh air. If exposed person is not breathing, give artificial respiration or oxygen applied by trained personnel. Get medical attention immediately.

Ingestion : If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

Most important symptoms/effects, acute and delayed

Eye contact : Causes serious eye damage.

Skin contact : Causes severe burns.

Inhalation : May give off gas, vapor or dust that is very irritating or corrosive to the respiratory system. Exposure to decomposition products may cause a health hazard. Serious effects may be delayed following exposure.

Ingestion : May cause burns to mouth, throat and stomach.

Protection of first-aiders : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

Notes to physician : In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.

See toxicological information (Section 11)

5. Fire-fighting measures

Suitable fire extinguishing media : Use water spray, fog or foam.

Specific hazards arising from the chemical : In a fire or if heated, a pressure increase will occur and the container may burst.

Hazardous thermal decomposition products : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
nitrogen oxides
metal oxide/oxides

Specific fire-fighting methods : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

6. Accidental release measures

Personal precautions : Initiate company's spill response procedures immediately. Keep people out of area. Put on appropriate personal protective equipment (see section 8). Do not touch or walk through spilled material.

Environmental precautions : Avoid contact of spilled material and runoff with soil and surface waterways.

6. Accidental release measures

Methods for cleaning up : Follow company's spill procedures. Keep people away from spill. Put on appropriate personal protective equipment (see section 8). Absorb/neutralize liquid material. Use a tool to scoop up solid or absorbed material and put into appropriate labeled container. Use a tool to scoop up solid or absorbed material and place into appropriate labeled waste container. Use a water rinse for final clean-up.

7. Handling and storage

Handling : Do not get in eyes or on skin or clothing. Do not breathe vapor or mist. Use only with adequate ventilation. Wash thoroughly after handling.

Storage : Keep out of reach of children. Keep container tightly closed. Keep container in a cool, well-ventilated area.

Do not store below the following temperature: 0°C

8. Exposure controls/personal protection

Control parameters

Ingredient name	Exposure limits
sodium hydroxide	**Taiwan Council of Labor Affairs (Taiwan, 1/2010).** STEL: 4 mg/m³ 15 minute(s). TWA: 2 mg/m³ 8 hour(s).

Biological indicator : Not available.

Appropriate engineering controls : Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Personal protection

Hygiene measures : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing.

Eye protection : Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.

Hand protection : Use chemical-resistant, impervious gloves.

Skin protection : Use synthetic apron, other protective equipment as necessary to prevent skin contact.

Respiratory protection : A respirator is not needed under normal and intended conditions of product use.

9. Physical and chemical properties

Appearance

Physical state : Liquid.

Color : Yellow [Light]

Odor : Odorless

pH : 13 to 14 (100%)

Melting point : Not available.

Boiling point : >100°C (>212°F)

Flash point : > 100°C

Evaporation rate (butyl acetate = 1) : Not available.

Flammability (solid, gas) : Not available.

9. Physical and chemical properties

Explosion limits : Not available.

Vapor pressure : Not available.

Vapor density : Not available.

Relative density : 1.15 to 1.28 (Water = 1)

Solubility : Easily soluble in the following materials: cold water and hot water.

Partition coefficient: n-octanol/water : Not available.

Auto-ignition temperature : Not available.

10. Stability and reactivity

Stability : The product is stable.

Possibility of hazardous reactions : Under normal conditions of storage and use, hazardous reactions will not occur.

Conditions to avoid : No specific data.

Materials to avoid : Extremely reactive or incompatible with the following materials: acids.
Slightly reactive or incompatible with the following materials: metals.

Hazardous decomposition products : Under normal conditions of storage and use, hazardous decomposition products should not be produced.

11. Toxicological information

Route of exposure : Skin contact, Eye contact, Inhalation, Ingestion

Symptoms

Eye contact : Adverse symptoms may include the following:
pain
watering
redness

Skin contact : Adverse symptoms may include the following:
pain or irritation
redness
blistering may occur

Inhalation : Adverse symptoms may include the following:
coughing
Respiratory tract irritation

Ingestion : Adverse symptoms may include the following:
stomach pains

Acute toxicity

Eye contact : Causes serious eye damage.

Skin contact : Causes severe burns.

Inhalation : May give off gas, vapor or dust that is very irritating or corrosive to the respiratory system. Exposure to decomposition products may cause a health hazard. Serious effects may be delayed following exposure.

Ingestion : May cause burns to mouth, throat and stomach.

Toxicity data

Product/ingredient name	Result	Species	Dose	Exposure
triethanolamine	LD50 Dermal	Rabbit	>2000 mg/kg	-
	LD50 Oral	Rat	6400 mg/kg	-
2,2'-iminodiethanol	LD50 Dermal	Rabbit	8180 mg/kg	-
	LD50 Oral	Rat	755 mg/kg	-

Chronic toxicity

Carcinogenicity : No known significant effects or critical hazards.

11. Toxicological information

Mutagenicity	:	No known significant effects or critical hazards.
Teratogenicity	:	No known significant effects or critical hazards.
Developmental effects	:	No known significant effects or critical hazards.
Fertility effects	:	No known significant effects or critical hazards.

12. Ecological information

Ecotoxicity : No known significant effects or critical hazards.

Aquatic and terrestrial toxicity

Product/ingredient name	Test	Result	Species	Exposure
sodium hydroxide	-	Acute EC50 40 mg/l	Daphnia	48 hours
triethanolamine	-	Acute LC50 11800 mg/l	Fish	96 hours
2,2'-iminodiethanol	-	Acute EC50 65.5 mg/l	Daphnia	48 hours

Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
Not available.				

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
Not available.			

Bioaccumulative potential

Product/ingredient name	$LogP_{ow}$	BCF	Potential
triethanolamine	-1	3.890451449	low
2,2'-iminodiethanol	-1.43	-	low

Mobility in soil : Not available.

Other adverse effects : No known significant effects or critical hazards.

13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Significant quantities of waste product residues should not be disposed of via the foul sewer but processed in a suitable effluent treatment plant. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

14. Transport information

Certain shipping modes or package sizes may have exceptions from the transport regulations. The classification provided may not reflect those exceptions and may not apply to all shipping modes or package sizes.

UN number	:	UN1824
Proper shipping name	:	SODIUM HYDROXIDE SOLUTION
Class	:	8
Packing group	:	II
Marine pollutant (Yes / No)	:	Not a pollutant.
Specific precautionary transport measures and conditions	:	-

14. Transport information

See shipping documents for specific transportation information.

15. Regulatory information

TCSCA List of toxic chemicals

Listed no.	Series no.	Product name	RQ	Class
Not available.				

List of chemicals for which manufacturing or handling is defined as "work specially hazardous to health" : Not available.

List of chemicals reputed to be a "threat of imminent danger" : This product contains substances considered to be a "Threat of imminent danger": sodium hydroxide.

LSHL Article 21 : Not available.

Safety, health and environmental regulations specific for the product : Not available.

References : -Regulation of Labeling and Hazard Communication of Dangerous and Harmful Materials.
-Labor Safety and Health Act.

16. Other information

History

Date of issue : **2012 . 12 . 20**

Date of previous issue : **2008 . 11 . 19**

Organization that prepared the SDS : ECOLAB

Person who prepared the SDS : Regulatory Affairs

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.